

31410 Hw 1

$$Q_1. \quad \begin{cases} \dot{x} = x + y + z \\ \dot{y} = y + z \\ \dot{z} = z \end{cases}$$

$$(a) \textcircled{1} \dot{z} = z \Rightarrow z(t) = z_0 e^t$$

$$\textcircled{2} \dot{y} = y + z \Rightarrow \ddot{y} = \dot{y} + \dot{z} \Rightarrow \ddot{y} = \dot{y} + z$$

$$\Rightarrow \ddot{y} = \dot{y} + (\dot{y} - y) = 2\dot{y} - y$$

$$\Rightarrow \ddot{y} - 2\dot{y} + y = 0$$

By solving the characteristic equation, we have

$$r^2 - 2r + 1 = 0 \Rightarrow (r-1)^2 = 0 \Rightarrow r=1 \text{ with multiplicity } 2.$$

$$\Rightarrow y(t) = (C_1 + C_2 t) e^t, \quad \dot{y}(t) = C_2 e^t + (C_1 + C_2 t) e^t$$

And since $y(0) = y_0$, $\dot{y}(0) = y_0 + z_0$. so

$$\begin{cases} y(0) = C_1 = y_0 \\ \dot{y}(0) = C_1 + C_2 = y_0 + C_2 = y_0 + z_0 \Rightarrow C_2 = z_0 \end{cases}$$

$$\Rightarrow y(t) = (y_0 + z_0 t) e^t$$

$\textcircled{3}$ Since $\dot{z} = z$ and $\dot{y} = y + z$, $y = \dot{x} - x - z$, so

$$\dot{y} = \ddot{x} - \dot{x} - \dot{z}, \quad \dot{y} = (\dot{x} - x - z) + z = \dot{x} - x$$

$$\Rightarrow \ddot{x} - \dot{x} - \dot{z} = \dot{x} - x \Rightarrow \ddot{x} - 2\dot{x} + x = z \Rightarrow \ddot{x} = 2\dot{x} - x + z$$

$$\Rightarrow \ddot{x} - 2\dot{x} + x = \dot{z} = z = \dot{x} - 2\dot{x} + x$$

$$\Rightarrow \ddot{x} - 3\dot{x} + 3x - x = 0$$

so the characteristic eqn is:

suppose $x(t) = e^{rt}$. then

$$r^3 e^{rt} - 3r^2 e^{rt} + 3r e^{rt} - e^{rt} = 0$$

$$\Rightarrow r^3 - 3r^2 + 3r - 1 = 0 \Rightarrow (r-1)^3 = 0 \Rightarrow r=1 \text{ with multiplicity } 3.$$

$$\Rightarrow x(t) = (A + Bt + Ct^2)e^t.$$

And since $x(0) = x_0$, $\dot{x}(0) = y_0 + z_0$, $\ddot{x}(0) = x_0 + 2y_0 + 3z_0$, so

$$x(0) = (A)e^0 = A = x_0.$$

$$x'(t) = (B + 2Ct)e^t + (A + Bt + Ct^2)e^t \Rightarrow x'(0) = B + A = x_0 + y_0 + z_0$$

$$x''(t) = (2C)e^t + (B + 2Ct)e^t + (B + 2Ct)e^t + (A + Bt + Ct^2)e^t$$

$$\ddot{x}(0) = 2C + B + B + \cancel{A} = 2C + x_0 + 2y_0 + 2z_0 = \cancel{x_0} + 2y_0 + 3z_0$$

$$\Rightarrow 2C = z_0 \Rightarrow C = \frac{1}{2}z_0$$

thus, $A = x_0$, $B = y_0 + z_0$, $C = \frac{1}{2}z_0$. so

$$x(t) = (x_0 + (y_0 + z_0)t + \frac{1}{2}z_0 t^2)e^t$$

Therefore, the three solutions are

$$\begin{cases} z(t) = z_0 e^t \\ y(t) = (y_0 + z_0 t)e^t, \\ x(t) = (x_0 + (y_0 + z_0)t + \frac{1}{2}z_0 t^2)e^t. \end{cases}$$

(b). still start with
$$\begin{cases} \dot{x} = x + y + z & ① \\ \dot{y} = y + z & ② \\ \dot{z} = z & ③ \end{cases}$$

From ③, $\dot{z} = z \Rightarrow z(t) = z_0 e^t$. ④

Take ④ into ②, we have

$$\begin{aligned} \dot{y} &= y + z_0 e^t \Rightarrow \dot{y} - y = z_0 e^t \\ &\Rightarrow e^{-t} \dot{y} - e^{-t} y = z_0 \\ &\Rightarrow \frac{d}{dt}(y e^{-t}) = z_0 \\ &\Rightarrow y e^{-t} = z_0 t + C_1 \\ &\Rightarrow y(t) = (C_1 + z_0 t) e^t \end{aligned}$$

Since $y(0) = y_0$, so $y(0) = C_1 = y_0 \Rightarrow y(t) = (y_0 + z_0 t) e^t$. ⑤

Take ⑤ and ④ to ①, we have

$$\begin{aligned} \dot{x} - x &= (y_0 + z_0 t) e^t + z_0 e^t = e^t (y_0 + z_0 (t+1)) \\ &\Rightarrow e^{-t} (\dot{x} - x) = y_0 + z_0 (t+1) \\ &\Rightarrow \frac{d}{dt}(x e^{-t}) = y_0 + z_0 (t+1) \\ &\Rightarrow x e^{-t} = \int (y_0 + z_0 (t+1)) dt = y_0 t + z_0 \left(\frac{t^2}{2} + t \right) + C_2 \\ &\Rightarrow x(t) = \left[y_0 t + C_2 + z_0 \left(\frac{t^2}{2} + t \right) \right] e^t \end{aligned}$$

And since $x(0) = C_2 = x_0$, so

$$x(t) = \left(x_0 + y_0 t + z_0 \left(\frac{t^2}{2} + t \right) \right) e^t.$$

(c) To rewrite (1) as a matrix form.

$$\dot{x} = Ax \Rightarrow \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$\Rightarrow$ the eigenvalues of A are $\lambda_1 = \lambda_2 = \lambda_3 = 1$. So the multiplicity is 3.

$$A - \lambda I = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow (A - \lambda I)\underline{v} = \underline{0} \Rightarrow v_3 = v_2 = 0, v_1 \in \mathbb{R}$$
$$\Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ is the eigenvector. } \underline{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

then to find the two independent generalized eigenvectors $\underline{w}$ and $\underline{u}$, we have

$$\begin{cases} (A - \lambda I)\underline{w} = \underline{v} \Rightarrow \underline{w} = [0, 1, 0]^T \\ (A - \lambda I)\underline{u} = \underline{w} \Rightarrow \underline{u} = [0, -1, 1]^T. \end{cases}$$

so we have

$$\underline{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \underline{w} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \underline{u} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \text{ and}$$

we have three independent solution of the system.

$$X_1(t) = e^t \underline{v}, \quad X_2(t) = e^t (t \underline{v} + \underline{w}), \quad X_3(t) = e^t \left(\frac{1}{2} t^2 \underline{v} + t \underline{w} + \underline{u} \right)$$

$$\Rightarrow X(t) = C_1 X_1(t) + C_2 X_2(t) + C_3 X_3(t)$$

$$= C_1 \begin{bmatrix} e^t \\ 0 \\ 0 \end{bmatrix} + C_2 \begin{bmatrix} te^t \\ e^t \\ 0 \end{bmatrix} + C_3 \begin{bmatrix} \frac{1}{2}t^2e^t \\ te^t - e^t \\ e^t \end{bmatrix}$$

$$= \begin{bmatrix} C_1 e^t + C_2 te^t + C_3 \frac{1}{2}t^2e^t \\ C_2 e^t + C_3 (te^t - e^t) \\ C_3 e^t \end{bmatrix}$$

$$\Rightarrow X(0) = \begin{bmatrix} C_1 + 0 + 0 \\ C_2 - C_3 \\ C_3 \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} \Rightarrow \begin{cases} C_1 = x_0 \\ C_2 = y_0 + z_0 \\ C_3 = z_0 \end{cases}$$

thus.

$$X(t) = x_0 e^t \underline{v} + (y_0 + z_0) e^t (t \underline{v} + \underline{w}) + z_0 \left(\frac{1}{2} t^2 \underline{v} + t \underline{w} + \underline{u} \right)$$

$$= \begin{bmatrix} e^t (x_0 + (y_0 + z_0)t + \frac{1}{2} z_0 t^2) \\ e^t (y_0 + z_0 t) \\ z_0 e^t \end{bmatrix}$$

$$\Rightarrow \begin{cases} X(t) = e^t (x_0 + (y_0 + z_0)t + \frac{1}{2} z_0 t^2) \\ Y(t) = e^t (y_0 + z_0 t) \\ Z(t) = z_0 e^t. \end{cases}$$

□.

$$(d) \dot{V} = AV, \quad A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad V(0) = V_0 = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$$

First we decompose A .

$$A = I + N, \text{ with } N = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow e^{At} = e^{(I+N)t} = e^t e^{Nt} \text{ since } [It, Nt] = t^2[I, N] = 0$$

then we expand e^{Nt} as a finite series. since N is nilpotent. so

$$N^3 = 0$$

$$\Rightarrow e^{Nt} = I + Nt + \frac{N^2 t^2}{2!}, \text{ with}$$

$$N^2 = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\text{so } e^{Nt} = \begin{pmatrix} 1 & t & t + \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow V(t) = e^{At} V_0 = e^t \begin{pmatrix} 1 & t & t + \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} x(t) = e^t (x_0 + y_0 t + z_0 (t + \frac{t^2}{2})) \\ y(t) = e^t (y_0 + t z_0) \\ z(t) = e^t z_0 \end{cases} \quad \square$$

(e). I think that approach (d) is the most elegant, it provides a compact expression for the entire solution and clearly shows the role of the Jordan block through the t and t^2 terms.

The most tedious one was (a). Deriving and solving the third-order ODE for $x(t)$ require long algebraic substitutions and offered little new insight compared with the cleaner matrix - exponential approach.

Q2. (a)

let's compute the det of M first.

let $a = -1 + \gamma + \gamma \cos 2t$, $b = 1 - \gamma \sin 2t$, $c = -1 - \gamma \sin 2t$, $d = -1 + \gamma - \gamma \cos 2t$.

then $\det(M) = ad - bc$

$$\begin{aligned} &= (-1 + \gamma + \gamma \cos 2t)(-1 + \gamma - \gamma \cos 2t) - (1 - \gamma \sin 2t)(-1 - \gamma \sin 2t) \\ &= \cancel{-1} + \gamma \cos 2t - \cancel{-1} + \gamma^2 - \gamma^2 \cos 2t - \gamma \cos 2t + \gamma^2 \cos 2t - \gamma^2 \cos^2 2t - (\cancel{1} - \gamma \sin 2t + \gamma \sin 2t + \gamma^2 \sin^2 2t) \\ &= 1 + \gamma^2 - 2\gamma - \gamma^2 \cos^2 2t - (1 - \gamma \sin 2t + \gamma \sin 2t + \gamma^2 \sin^2 2t) \\ &= \cancel{1} + \gamma^2 - \cancel{2\gamma} - \gamma^2 \cos^2 2t + \cancel{1} + \gamma \cancel{\sin 2t} - \gamma \cancel{\sin 2t} - \gamma^2 \sin^2 2t \\ &= 1 + \gamma^2 - 2\gamma + 1 - \gamma^2 = 2 - 2\gamma \end{aligned}$$

so the determinant is independent of t .

then, for the eigenvalues.

$$\begin{aligned} \lambda_{1,2} &= \frac{\text{tr}(M) \pm \sqrt{(\text{tr} M)^2 - 4 \det M}}{2} \\ &= \frac{-2 + 2\gamma \pm \sqrt{(2\gamma - 2)^2 - 4(2 - 2\gamma)}}{2} \\ &= \frac{-2 + 2\gamma \pm \sqrt{4\gamma^2 - 8\gamma + 4 - 8 + 8\gamma}}{2} = \frac{-2 + 2\gamma \pm \sqrt{4\gamma^2 - 4}}{2} \\ &= \gamma - 1 \pm \sqrt{\gamma^2 - 1}, \text{ which are constant in time.} \end{aligned}$$

To classify the stability.

$$\lambda_{1,2} = \gamma - 1 \pm i\sqrt{1 - \gamma^2}$$

when $-1 \leq \gamma < 1$, then $\gamma - 1 < 0 \Rightarrow X = 0$ is asymptotically stable

when $\gamma = 1$. $\lambda_1 = \lambda_2 = 0$, $\Rightarrow X=0$ is a center, which is Lyapunov stable

when $\gamma > 1$, $\gamma - 1 + \sqrt{\gamma^2 - 1} > 0$. $\gamma - 1 - \sqrt{\gamma^2 - 1} < 0 \Rightarrow X=0$ is a saddle point unstable

when $\gamma < -1$, $\lambda_1 < 0$, $\lambda_2 < 0 \Rightarrow X=0$ is asymptotically stable.

(b) Let's verify first that $X_1(t)$ satisfies $\dot{X} = M(t)X$.

$$\begin{aligned}\dot{X}_1 &= \begin{pmatrix} -\sin t \\ -\cos t \end{pmatrix} e^{(2r-1)t} + (2r-1) \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} e^{(2r-1)t} \\ &= e^{(2r-1)t} \begin{pmatrix} (2r-1)\cos t - \sin t \\ -(2r-1)\sin t - \cos t \end{pmatrix}\end{aligned}$$

$$M(t)X = \begin{pmatrix} -1+r+r\cos 2t & 1-r\sin 2t \\ -1-r\sin 2t & -1+r-r\cos 2t \end{pmatrix} \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} e^{(2r-1)t}$$

$$\begin{aligned}& (-1+r+r\cos 2t) \cos t - \sin t (1-r\sin 2t) \\ &= (-1+r+r(\cos^2 t - \sin^2 t)) \cos t - \sin t (1-2r\sin t \cos t) \sin t \\ &= (-1+r) \cos t + r\cos^3 t - r\sin^2 t \cos t - \sin t + 2r\sin^2 t \cos t \\ &= (-1+r) \cos t - \sin t + r\cos^3 t + r\sin^2 t \cos t \\ &= (-1+r) \cos t - \sin t + r\cos t (\cos^2 t + \sin^2 t) \\ &= (-1+r) \cos t - \sin t + r\cos t = (2r-1) \cos t - \sin t \\ & (-1-r\sin 2t) \cos t - (-1+r-r\cos 2t) \sin t \\ &= -\cos t - r\sin 2t \cos t + \sin t - r\sin t + r\cos 2t \sin t \\ &= -\cos t + \sin t - r2\sin t \cos^2 t - r\sin t + r(\cos^2 t - \sin^2 t) \sin t \\ &= -\cos t + \sin t - 2r\sin t \cos^2 t - r\sin t + r\sin t \cos^2 t - r\sin^3 t \\ &= -\cos t + \sin t - r\sin t (\cos^2 t + \sin^2 t) - r\sin t = -\cos t - (2r-1) \sin t.\end{aligned}$$

thus.

$$M(t)X_1(t) = \begin{pmatrix} (2r-1)\cos t - \sin t \\ -\cos t - (2r-1)\sin t \end{pmatrix} e^{(2r-1)t} = \dot{X}_1(t)$$

$\Rightarrow X_1(t)$ is indeed a solution of (2).

then let's verify $X_2(t)$ is indeed a soln to (2).

$$\dot{X}_2(t) = \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} e^{-t} + \begin{pmatrix} \sin t \\ \cos t \end{pmatrix} (-e^{-t})$$

$$= e^{-t} \begin{pmatrix} \cos t - \sin t \\ -\sin t - \cos t \end{pmatrix}$$

$$M(t)X_2 = \begin{pmatrix} -1+r+r\cos(2t) & 1-r\sin(2t) \\ -1-r\sin(2t) & -1+r-r\cos(2t) \end{pmatrix} \begin{pmatrix} \sin t \\ \cos t \end{pmatrix} e^{-t}$$

$$(-1+r+r\cos 2t)\sin t + (1-r\sin 2t)\cos t$$

$$= (r-1)\sin t + r(\cos^2 t - \sin^2 t)\sin t + \cos t - r(2\sin t \cos^2 t)$$

$$= (r-1)\sin t + \cos t - r\sin^3 t - r\sin t \cos^2 t$$

$$= (r-1)\sin t + \cos t - r\sin t(\cos^2 t + \sin^2 t) = (r-1)\sin t + \cos t - r\sin t$$

$$= -\sin t + \cos t$$

$$(-1 - \gamma \sin 2t) \sin t + (-1 + \gamma - \gamma \cos 2t) (\cos t)$$

$$= -\sin t - 2\gamma \sin^2 t \cos t - \cos t + \gamma \cos t - \gamma (\cos^3 t - \cos t \sin^2 t)$$

$$= -\sin t - \cos t + \gamma \cos t - \gamma \cos^3 t + \gamma \cos t \sin^2 t - 2\gamma \sin^2 t \cos t$$

$$= -\sin t - \cos t + \gamma \cos t - \gamma \cos t (\cos^2 t + \sin^2 t) = -\sin t - \cos t$$

$$\Rightarrow \dot{X}_2(t) = M(t)X_2(t) \Rightarrow X_2(t) \text{ is indeed a solution to (2)}$$

to show linear independent. we need to verify $[X_1 \ X_2]$ is invertible

$$\begin{vmatrix} \cos t e^{(2\gamma-1)t} & \sin t e^{-t} \\ -\sin t e^{(2\gamma-1)t} & \cos t e^{-t} \end{vmatrix} = e^{2(\gamma-1)t} \neq 0$$

$\Rightarrow X_1(t)$ and $X_2(t)$ are linearly independent.

And since initial condition is $X(0) = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$, so

$$X(t) = x_0 e^{(2\gamma-1)t} \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} + y_0 e^{-t} \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$$

when $2\gamma-1 > 0$, i.e. $\gamma > \frac{1}{2}$, then as times goes by, $X(t)$ will be far and far away from 0.

when $\gamma = \frac{1}{2}$, then if $x_0 \neq 0$, then as time goes by, $X(t)$ will be closer and closer to the set $\{(x, y) : x^2 + y^2 = x_0^2\}$.

when $\gamma < \frac{1}{2}$, then as $t \rightarrow \infty$, $X(t) \rightarrow \underline{0}$.

$$P e^{At} P^{-1} X(0).$$

If $\gamma = 0$, then we have $M(t) = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$, which is a constant matrix such that all the solutions of (2) can be expressed in terms of the eigenvalues / eigenvectors of M .

$$e^{Mt} x_0 = \left(I + Mt + \frac{1}{2} M^2 t^2 + \dots \right) x_0$$

If $\gamma \neq 0$, $M(t)$ is π -periodic, according to Theorem 2.36 in Book.

Let M be the monodromy matrix for a T -periodic linear system $\dot{x} = A(t)x$ and $TB = \ln M$. Then there exists a T -periodic matrix P s.t. the fundamental solution is $\Phi(t, 0) = P e^{tB}$

Therefore, when $\gamma \neq 0$, all soln of (2) are expressed in terms of the fundamental matrix solution of (2) rather than the eigenvec / eigval of M .

(c), The definition of fundamental matrix solution should be

$$\dot{\Phi}(t) = M(t)\Phi(t), \det(\Phi(t)) \neq 0, \forall t$$

And from (b) we know that

$$\begin{cases} \dot{x}_1(t) = M(t)x_1(t) \\ \dot{x}_2(t) = M(t)x_2(t) \end{cases} \Rightarrow \begin{bmatrix} \dot{x}_1(t) & \dot{x}_2(t) \end{bmatrix} = M(t) \begin{bmatrix} x_1(t) & x_2(t) \end{bmatrix}$$

And we show that $\begin{vmatrix} e^{(2\gamma-1)t} \cos t & e^{-t} \sin t \\ -\sin t e^{(2\gamma-1)t} & e^{-t} \cos t \end{vmatrix} = 1 \neq 0, \forall t$

$\Rightarrow$ the fundamental solution is $\Phi(t) = \begin{bmatrix} e^{(2\gamma-1)t} \cos t & e^{-t} \sin t \\ -\sin t e^{(2\gamma-1)t} & e^{-t} \cos t \end{bmatrix}$

its initial condition is

$$\Phi(\omega) = \begin{bmatrix} e^0 \cos 0 & e^0 \sin 0 \\ -\sin 0 e^0 & e^0 \cos 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \text{Id}_{2 \times 2}.$$

Because $M(t)$ contains $\sin(2t)$ and $\cos(2t)$, its period is

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi$$

For the monodromy matrix,

$$\begin{aligned} \Phi(T) &= \Phi(\pi) \\ \Rightarrow \Phi(\pi) &= \begin{pmatrix} \cos \pi & \sin \pi \\ -\sin \pi & \cos \pi \end{pmatrix} \begin{pmatrix} e^{(2r-1)\pi} & 0 \\ 0 & e^{-\pi} \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} e^{(2r-1)\pi} & 0 \\ 0 & e^{-\pi} \end{pmatrix} \end{aligned}$$

Hence,

$$\Phi(\pi) = -\text{diag}(e^{(2r-1)\pi}, e^{-\pi}) = \text{diag}(-e^{(2r-1)\pi}, -e^{-\pi}).$$

since the Floquet multipliers are just the eigenvalues of $\Phi(T) = \Phi(\pi)$.

so

$$\mu_1 = -e^{(2r-1)\pi}, \quad \mu_2 = -e^{-\pi},$$

which are two Floquet multipliers for this problem

(the rest of Q2 (c) (d) (e) and Q3 are in the latex file)

Q2(C) & Q2(D) & Q2(E)

Now let's apply Floquet theory to analyze the solution of (2):

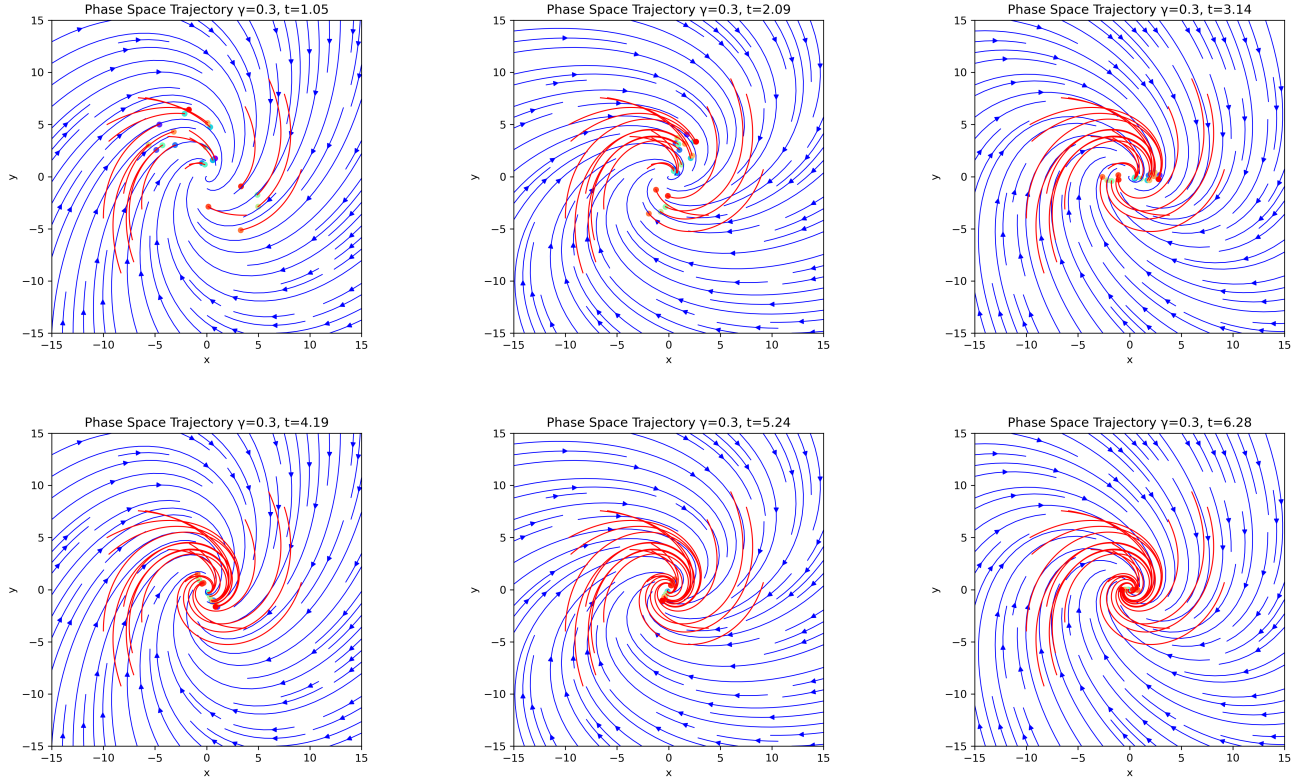


FIGURE 1. Phase portrait of (2) with $\gamma = 0.3$

- If $|\mu_1| < 1 \iff \gamma < \frac{1}{2}$, we have $\forall \mathbf{X}(0) \in \mathbb{R}^2$, $\mathbf{X}(t) \rightarrow \mathbf{0}$ as $t \rightarrow \infty$. Because

$$\lim_{t \rightarrow \infty} \mathbf{X}(t) = \lim_{n \rightarrow \infty} \mathbf{X}(nT) = \lim_{n \rightarrow \infty} M^n \mathbf{X}_0 = \begin{pmatrix} -e^{(2\gamma-1)n\pi} & 0 \\ 0 & -e^{-n\pi} \end{pmatrix} \mathbf{X}_0 = \mathbf{0}.$$

So here $\mathbf{0}$ is a globally asymptotically stable equilibrium of (2). To intuitively show this, I plot the phase portrait of (2) with $\gamma = 0.3$ below (see Figure 1), where I randomly set 20 initial conditions in the square $[-10, 10] \times [-10, 10]$, and took the snapshots when $t = \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}, 2\pi$. In these graphs, the red curves are the trajectory of the particles for time from 0 to t , blue curves are the vector fields at time t , and the points with different colors are the positions of the particles at time t . We can intuitively find that as time goes by, all particles are gradually drawn into 0, which agrees with the conclusion we have drawn above and in (b).

- If $|\mu_1| = 1$, then $\gamma = \frac{1}{2}$, for $\mathbf{X}_0 = (x_0, y_0)^\top$,

$$\mathbf{X}(nT) = M^n \mathbf{X}_0 = (x_0, -e^{-n\pi} y_0)^\top.$$

We can take some snapshots to see the behavior of the particles as below (see Figure 2). Combining the results of the snapshots and the fundamental solution matrix $\Phi(t)$, we can see that as time goes by, the motion of almost all particles ($x_0 \neq 0$) will approach circular motion. Therefore, the equilibrium point 0 is Lyapunov stable, which is consistent with the conclusion we have drawn above and in (b), and contradicts the conclusion we have drawn in (a).

- If $|\mu_1| > 1 \iff \gamma > \frac{1}{2}$, according to the Floquet theory, for $\mathbf{X}_0 = (x_0, y_0)^\top$, we have

$$\mathbf{X}(nT) = M^n \mathbf{X}_0 = \begin{pmatrix} -e^{(2\gamma-1)n\pi} & 0 \\ 0 & -e^{-n\pi} \end{pmatrix} \mathbf{X}_0 = (-e^{(2\gamma-1)n\pi} x_0, -e^{-n\pi} y_0)^\top.$$

we can see that as time goes by, all most all the particles ($x_0 \neq 0$) will go far and far away from 0, which means that the equilibrium point 0 is unstable. To intuitively show this, I plot the phase portrait of (2) with $\gamma = 0.8$ below (see Figure 3), where I randomly set 20 initial conditions in the square $[-10, 10] \times [-10, 10]$,

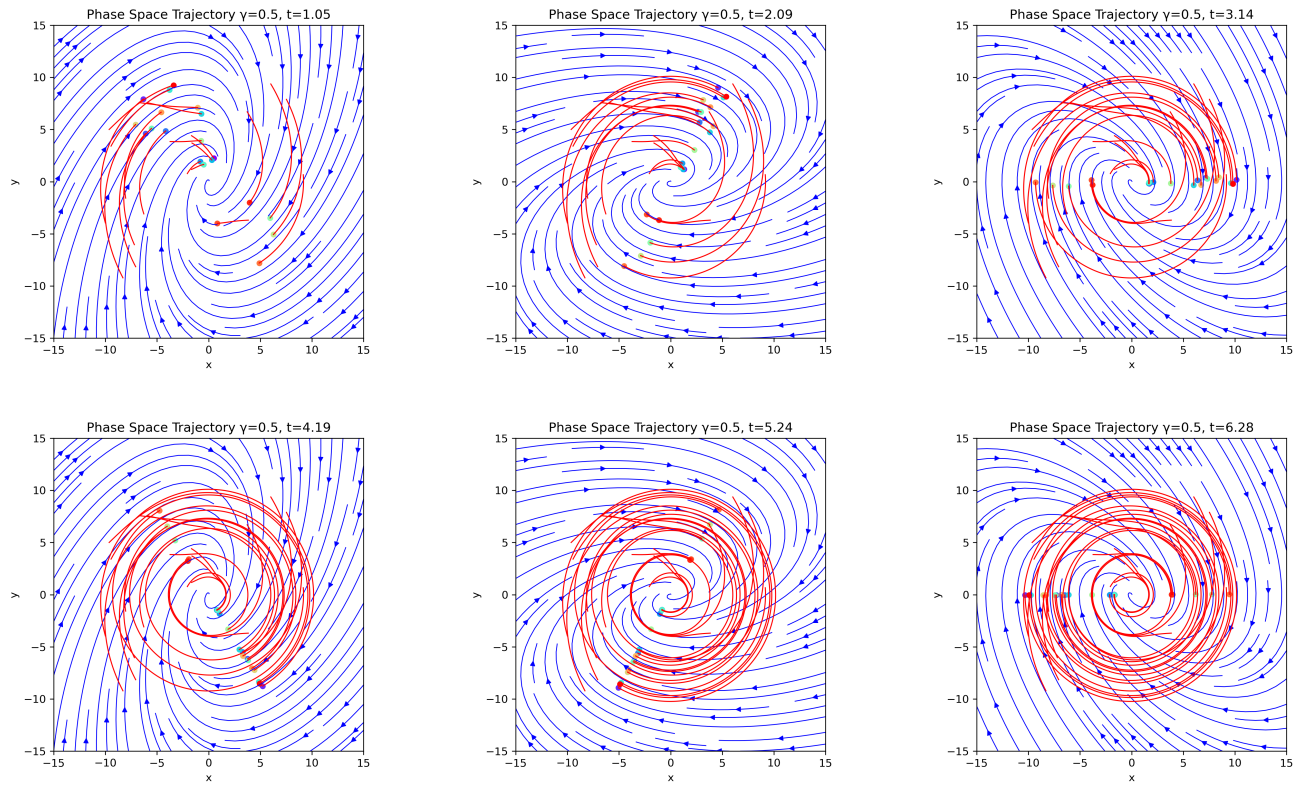


FIGURE 2. Phase portrait of (2) with $\gamma = 0.5$

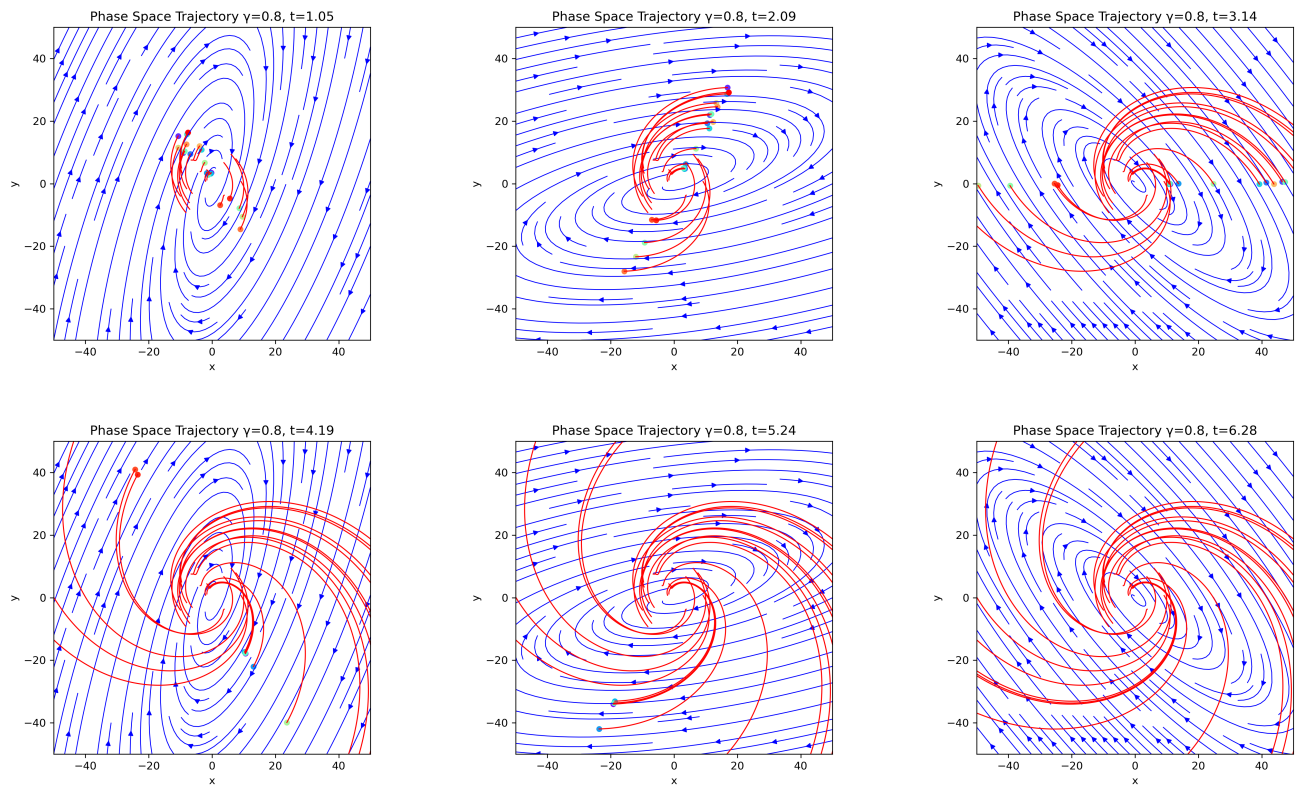


FIGURE 3. Phase portrait of (2) with $\gamma = 0.8$

and took the snapshots when $t = \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}, 2\pi$. In these graphs, the red curves are the trajectory of the particles for time from 0 to t , blue curves are the vector fields at time t , and the points with different colors are the positions of the particles at time t . We can intuitively find that as time goes by, almost all particles are far and far away from 0, which agrees with the conclusion we have drawn above and in (b), and contradicts the conclusion we have drawn in (a).

Conclusion:

Based on what I have learned in parts (a) and (b), I found that if I don't use the Floquet theory to deal with the linear, homogeneous differential equations with periodic matrix, I will get the wrong conclusion, which means there is a huge different between the autonomous and periodic systems. Therefore, it's very important to apply the Floquet theory in the study of linear, homogeneous differential equations with periodic matrix.

For which values of γ do all solutions approach $\mathbf{X} = 0$ as $t \rightarrow \infty$:

From the analysis above, we know that when $\gamma < \frac{1}{2}$, all solutions approach $\mathbf{X} = 0$ as $t \rightarrow \infty$.

(d) From the definition of $M(t)$, we know that its period is $T = \pi$. I plot the phase portrait of system (2) with $\gamma = 0.75$ and $t = \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{5\pi}{6}, \pi$ below (see Figure 4). In each panel, the **blue streamlines** represent

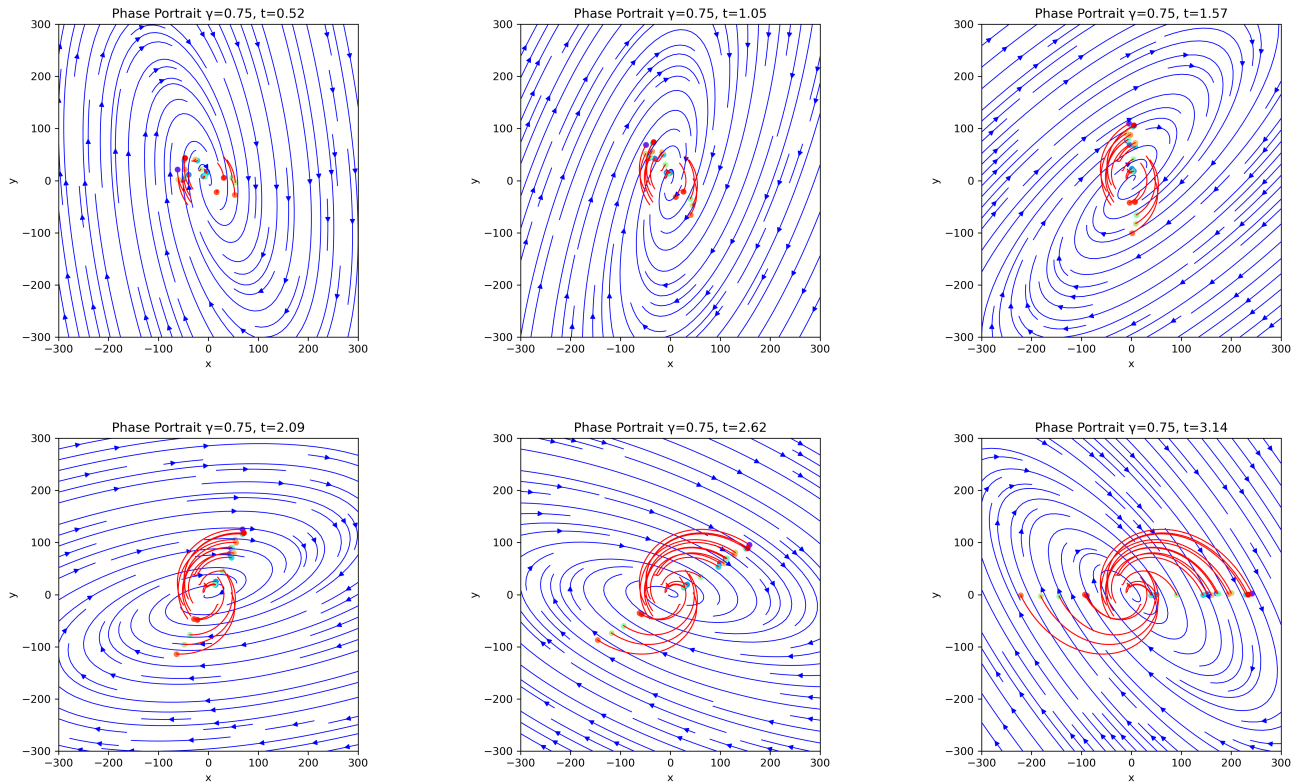


FIGURE 4. Phase portrait of (2) with $\gamma = 0.75$

the instantaneous vector field $\dot{X} = M(t_j)X$ at time t_j . To illustrate the system's behavior under various initial conditions, I randomly choose **20 initial points** uniformly distributed in the square $[-300, 300] \times [-300, 300]$. The **red curves** denote the trajectories of these particles, each integrated from $t = 0$ up to the corresponding time t_j , while the **colored dots** mark their current positions at time t_j .

As predicted by Floquet theory, since $\gamma = 0.75 > 0.5$, the Floquet multiplier satisfies $|\mu_1| > 1$. Consequently, all trajectories spiral outward from the origin as time increases, indicating that $X = 0$ is an **unstable spiral (repelling focus)**. The instantaneous vector fields slightly rotate over one period due to the time-periodic nature of $M(t)$, but the overall dynamics remain dominated by exponential divergence away from the origin.

To answer the mechanism of why a solution diverging to infinity even though we might have thought, based on the snapshots of the phase portrait, that solutions are spiraling into the origin, let's see that, for any fixed time t_0 , we obtain a constant matrix $M(t_0)$ whose instantaneous phase portrait appears as a spiral into the origin. By analyzing the eigenvalues of $M(t_0)$, we find that for $\gamma \in (-1, 1)$ each frozen-time system $\dot{X} = M(t_0)X$ is locally

asymptotically stable, so according to the conclusion in part (a), one might expect $X = 0$ to be stable. However, this intuition is misleading because the vector field $M(t)$ is changing continuously with time. The directions of contraction and expansion rotate as t evolves, so that even though the system appears to be spiraling inward at every frozen snapshot, the cumulative effect over one full period can produce amplification rather than decay. From Floquet theory, when $\gamma > \frac{1}{2}$, the Floquet multiplier satisfies $|\mu_1| > 1$, which means that any nonzero solution will eventually grow without bound as $t \rightarrow \infty$. In simple terms, the time-periodic changes in $M(t)$ cause the system to behave like a *parametrically excited oscillator*. Although each snapshot of the vector field looks stable, the directions of stretching and compression keep rotating over time. As a result, small amplifications that occur at certain phases accumulate from one period to the next, producing an overall exponential growth of trajectories. This explains why the solutions diverge even though every instantaneous phase portrait seems to show a stable spiral. Therefore, when dealing with time-dependent systems, we must rely on Floquet analysis instead of trusting our intuition from static phase portraits.

If we reversing the time in (2), we have

$$\frac{d\mathbf{X}}{dt} = \mathbf{M}(-t)\mathbf{X}, \mathbf{X} \in \mathbb{R}^2.$$

Still keep the initial condition $\mathbf{X}(0) = (1, 1)^T$ and $\gamma = 0.75$, I plotted the phase portraits of the reversed-time system at six equally spaced time steps from $t = 0$ to $t = 4\pi$, as shown in Figure 5.

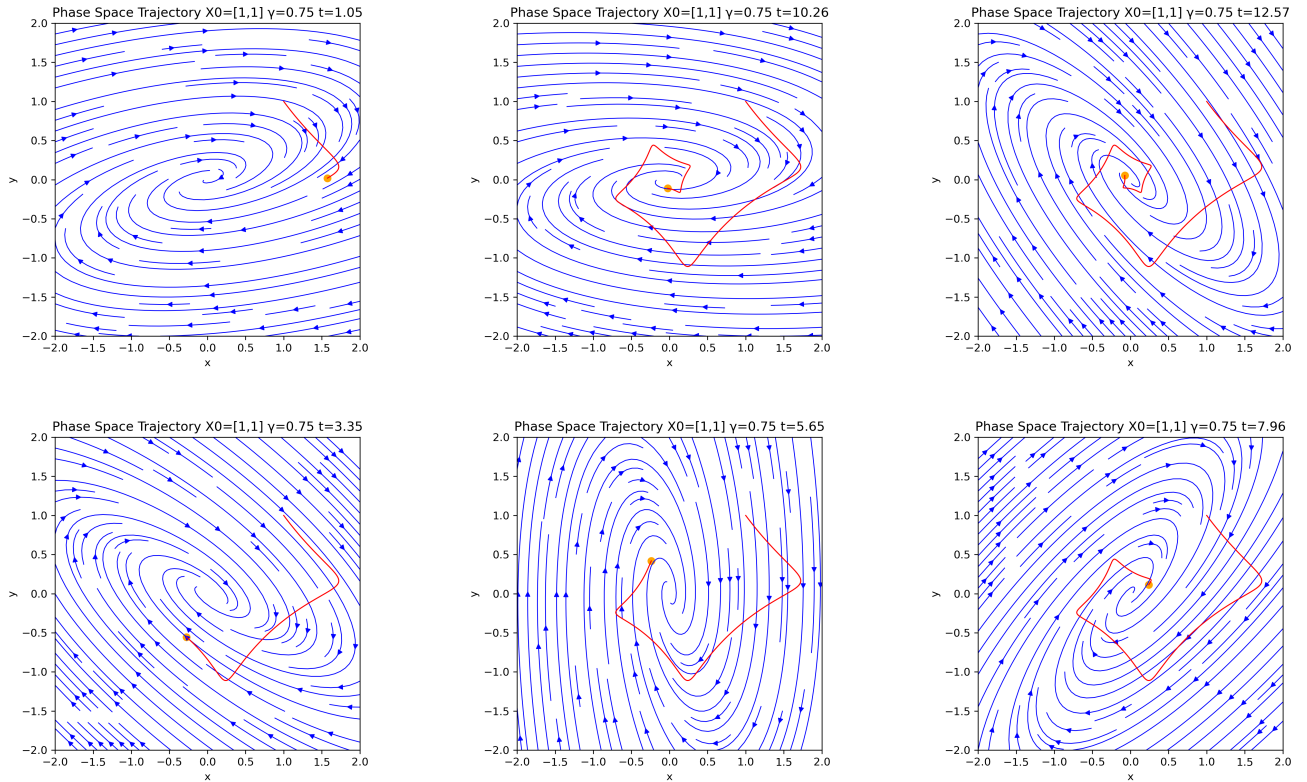


FIGURE 5. Phase portrait with reversed time

From the snapshots above, we can see that the particles are spiraling into the origin along a very weird trajectory.

To further investigate how the parameter γ influences the system stability, I plotted the phase portraits of (2) for $\gamma = 0.7, 0.8, 0.9, 1.0, 1.1$, and 1.2 , as shown in Figure 7. Each panel displays the vector field at $t = 2\pi$ and the trajectory of a particle starting from the initial condition $X(0) = (1, 1)^T$. The results clearly show a gradual transition from stable to unstable behavior as γ increases.

For $\gamma < 1$, the trajectories spiral inward toward the origin, indicating that the equilibrium point $X = 0$ is asymptotically stable. When $\gamma = 1$, the system lies approximately at the boundary between stability and instability, where the trajectory neither grows nor decays significantly. For $\gamma > 1$, the trajectories spiral outward and their

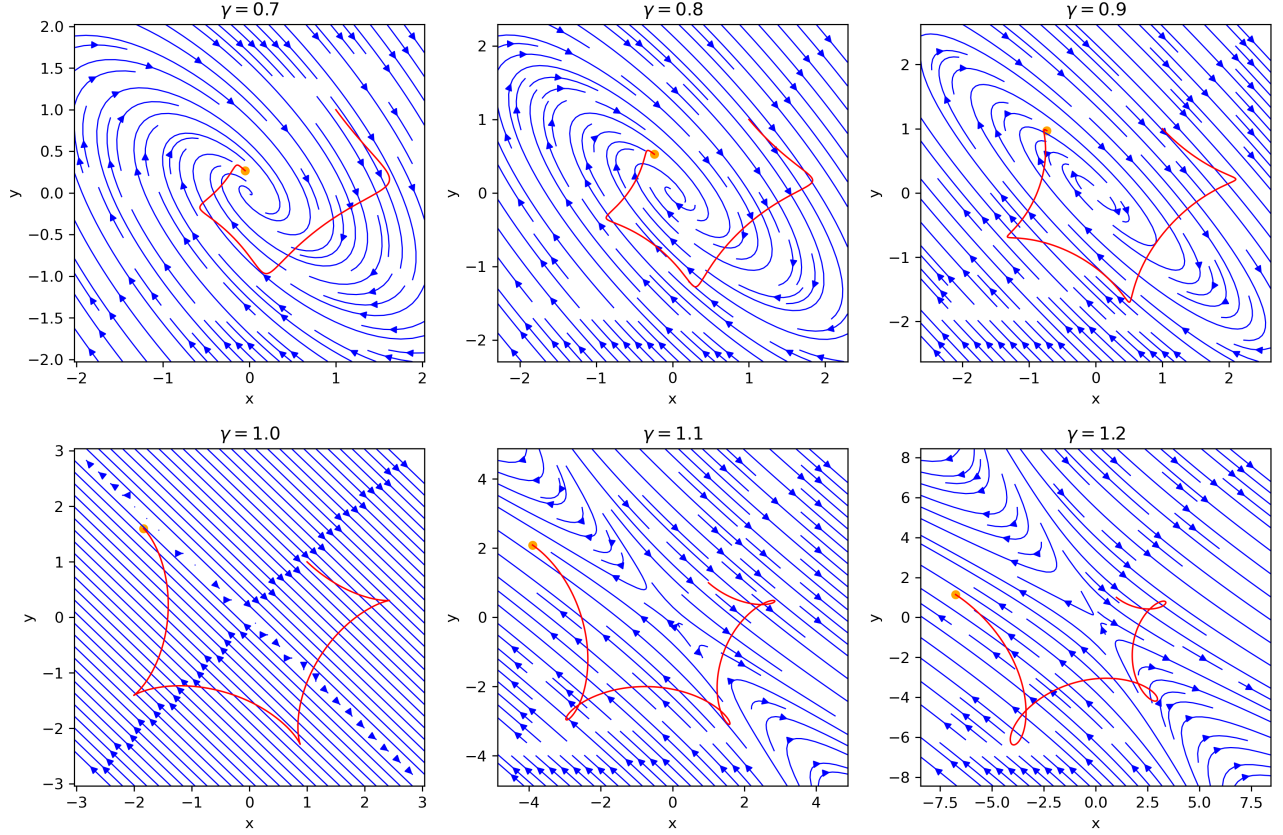


FIGURE 6. Phase portraits of (2) for $\gamma = 0.7, 0.8, 0.9, 1.0, 1.1$, and 1.2

amplitudes increase over time, revealing that $X = 0$ becomes an unstable spiral.

(e) Proposed extension.

To further extend the analysis, we may start from Abel's Theorem, which provides a relationship between the trace of the coefficient matrix and the determinant of the fundamental matrix solution. For a linear time-dependent system

$$\dot{X} = M(t)X,$$

the determinant of its fundamental solution $\Phi(t)$ satisfies

$$\frac{d}{dt} \det(\Phi(t)) = \text{tr}(M(t)) \det(\Phi(t)).$$

Integrating from 0 to one period T , we obtain

$$\det(\Phi(T)) = \exp\left(\int_0^T \text{tr}(M(s)) ds\right).$$

For our specific system,

$$M(t) = \begin{pmatrix} -1 + \gamma + \gamma \cos(2t) & 1 - \gamma \sin(2t) \\ -1 - \gamma \sin(2t) & -1 + \gamma - \gamma \cos(2t) \end{pmatrix},$$

we have $\text{tr}(M(t)) = 2(\gamma - 1)$, which is constant in t . Therefore,

$$\det(\Phi(T)) = e^{2\pi(\gamma-1)}.$$

This expression gives the product of the two Floquet multipliers,

$$\mu_1 \mu_2 = e^{2\pi(\gamma-1)},$$

and implies that the overall phase-space volume expands when $\gamma > 1$ and contracts when $\gamma < 1$.

However, this result alone cannot determine the individual values of μ_1 and μ_2 , and thus does not fully characterize the stability of the equilibrium $X = 0$. In particular, for the reversed-time system

$$\dot{X} = M(-t)X,$$

although $\text{tr}(M(-t)) = \text{tr}(M(t))$ and hence $\det(\Phi(T)) = e^{2\pi(\gamma-1)}$ remains the same, we still do not know the explicit form of the fundamental solution $\Phi(t)$, nor can we directly compute the corresponding monodromy matrix and its eigenvalues (Floquet multipliers). Therefore, we cannot rigorously determine whether $\lambda = 1$ still serves as the stability threshold for $X = 0$ under time reversal.

A natural extension of this problem would be to explicitly derive the fundamental solution for the reversed-time system, compute its monodromy matrix and Floquet multipliers, and verify whether the resulting analytical stability criterion matches the numerical observations obtained in part (d). This would provide a direct theoretical confirmation of how time reversal affects the stability and spectral properties of periodic linear systems.

Q3.

Throughout parts (a)–(d) we set $\varepsilon = 0$ so that

$$f(x, y) = (x - a)^2 + (y - b)^2.$$

It is convenient to write $z := (x, y)^\top$ and $z^* := (a, b)^\top$, so

$$f(z) = \|z - z^*\|_2^2, \quad \nabla f(z) = 2(z - z^*), \quad \nabla^2 f(z) = 2I_2.$$

(a) Unique minimizer for $\varepsilon = 0$. Since the Hessian is the constant positive definite matrix $\nabla^2 f \equiv 2I_2 \succ 0$, the function f is strictly convex on $\mathbb{R}^2$. The first-order optimality condition

$$\nabla f(z) = 0 \iff 2(z - z^*) = 0 \iff z = z^*$$

has the unique solution $z^* = (a, b)^\top$. Therefore,

$$\min_{z \in \mathbb{R}^2} f(z) = f(z^*) = 0, \quad \text{and the minimizer is unique.}$$

(b) Gradient descent as a contraction mapping. For $\varepsilon = 0$, we have

$$\nabla f(x, y) = \begin{pmatrix} 2(x - a) \\ 2(y - b) \end{pmatrix}.$$

The gradient–descent iteration with step size $\eta > 0$ is therefore

$$\begin{pmatrix} x_{k+1} \\ y_{k+1} \end{pmatrix} = \begin{pmatrix} x_k \\ y_k \end{pmatrix} - \eta \begin{pmatrix} 2(x_k - a) \\ 2(y_k - b) \end{pmatrix} = \begin{pmatrix} (1 - 2\eta)x_k + 2\eta a \\ (1 - 2\eta)y_k + 2\eta b \end{pmatrix}.$$

Define the update map

$$T_\eta(x, y) = ((1 - 2\eta)x + 2\eta a, (1 - 2\eta)y + 2\eta b).$$

To check whether T_η is a contraction in the Euclidean norm, take any two points (x_1, y_1) and (x_2, y_2) and compute

$$\|T_\eta(x_1, y_1) - T_\eta(x_2, y_2)\|_2 = |1 - 2\eta| \|(x_1 - x_2, y_1 - y_2)\|_2.$$

Hence T_η is a contraction if and only if its Lipschitz constant $c = |1 - 2\eta|$ satisfies $c < 1$, i.e.

$$|1 - 2\eta| < 1 \iff 0 < \eta < 1.$$

For any step size η in this interval, T_η is a contraction mapping on $\mathbb{R}^2$ and therefore the gradient–descent iteration converges to the unique fixed point $(x, y) = (a, b)$.

(c) Newton’s method converges in one step. Newton’s iteration is

$$z_{k+1} = z_k - [\nabla^2 f(z_k)]^{-1} \nabla f(z_k) = z_k - \frac{1}{2} \cdot 2(z_k - z^*) = z^*.$$

Thus *for any* initial guess z_0 , one Newton step lands exactly at the minimizer z^* . In particular, Newton’s method is globally convergent in a single iteration for this quadratic.

(d) Is Newton's update a contraction map? The Newton update map for this quadratic is the constant map

$$T_N(z) = z^* \quad \text{for all } z \in \mathbb{R}^2.$$

Hence for any u, v ,

$$\|T_N(u) - T_N(v)\|_2 = \|z^* - z^*\|_2 = 0 \leq c \|u - v\|_2 \quad \text{with } c = 0 < 1.$$

Formally, T_N is a contraction map with Lipschitz constant 0, and its unique fixed point is z^* . Conceptually, this simply reflects the fact that Newton's method for a strictly convex quadratic with constant Hessian reaches the solution in a single step, so the contraction-mapping framework does not provide a meaningful rate beyond that immediate convergence.

(e) Perturbed Objective and the Location of the Minimizer. (Discussed with Jingyi Zhu)

$$\begin{aligned} f(x, y) &= (x - a)^2 + (y - b)^2 + \varepsilon n(x, y) \\ &= (x - a)^2 + (y - b)^2 + 0.1 \sin(2\pi x) \cos(2\pi y) \\ &= (x - a)^2 + (y - b)^2 + 0.05 \sin(4\pi x) \end{aligned}$$

Notice that the term $(y - b)^2$ reaches its minimum when $y = b$, so the optimization can be reduced to a one-dimensional problem in x :

$$g(x) = (x - a)^2 + 0.05 \sin(4\pi x)$$

Taking the derivative gives

$$g'(x) = 2(x - a) + 0.2\pi \cos(4\pi x) = 0,$$

which leads to

$$x - a = -0.1\pi \cos(4\pi x).$$

This is a transcendental equation that admits multiple roots. Since the right-hand side is small in magnitude, we expect the minimizer to be located near $x = a$. Let $x = a + \delta$ with $|\delta|$ small. Substituting into the equation yields

$$\delta = -0.1\pi \cos(4\pi a + 4\pi\delta).$$

For sufficiently small δ , the term $4\pi\delta$ inside the cosine can be neglected, giving

$$\delta \approx -0.1\pi \cos(4\pi a).$$

Hence, an approximate expression for the minimizer of $f(x, y)$ is

$$(x^*, y^*) \approx (a - 0.1\pi \cos(4\pi a), b),$$

and the corresponding minimum value is roughly

$$f_{\min} \approx 0.01\pi^2 \cos^2(4\pi a) + 0.05 \sin[4\pi(a - 0.1\pi \cos(4\pi a))],$$

which is a number very close to zero.

(f) Is Newton's method still a contraction map? For $\varepsilon = 0.1$, the objective function

$$f(x, y) = (x - a)^2 + (y - b)^2 + \varepsilon \sin(2\pi x) \cos(2\pi y)$$

is no longer globally convex. Stationary points satisfy

$$\begin{cases} 2(x - a) + 2\pi\varepsilon \cos(2\pi x) \cos(2\pi y) = 0, \\ 2(y - b) - 2\pi\varepsilon \sin(2\pi x) \sin(2\pi y) = 0. \end{cases}$$

When $\varepsilon = 0$, the unique stationary point is $(x^*, y^*) = (a, b)$. For $\varepsilon > 0$, we may have additional local minima, I will show this by plotting plotting the objective function for a few values of a .

By examining the contour plot of $f(x, y)$ for $a = 0.25$, we observe that the central low-value region is not perfectly circular but instead exhibits several small depressions separated by shallow ridges. Each of these depressions corresponds to a distinct local minimum of f . This indicates that the quadratic bowl has been perturbed by the sinusoidal term into multiple small basins of attraction. In contrast, when $a = 0$ the contours are nearly circular, revealing a single global minimum. Therefore, for certain values of a , $f(x, y)$ possesses multiple local minima, and the Newton map is only locally contractive near each minimum but not globally over $\mathbb{R}^2$.

Because of these additional local minima, Newton's method

$$T(z) = z - [\nabla^2 f(z)]^{-1} \nabla f(z)$$

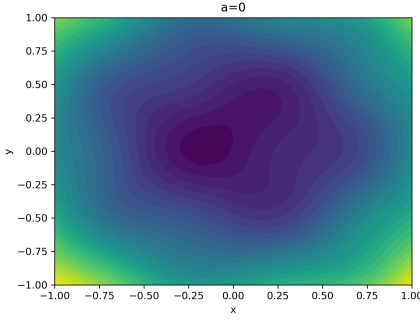
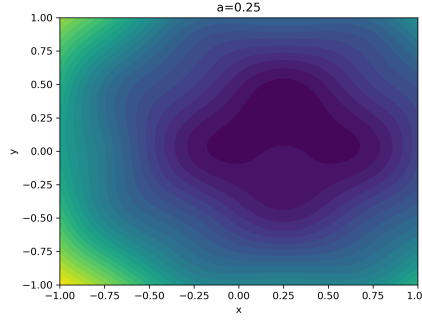
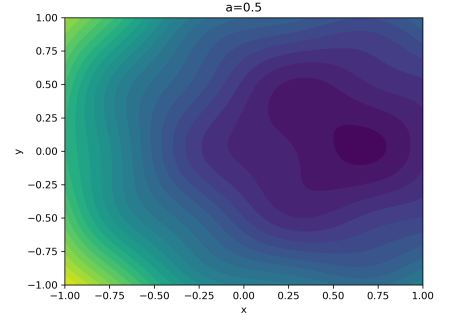
FIGURE 7. $a = 0$ FIGURE 8. $a = 0.25$ FIGURE 9. $a = 0.5$

FIGURE 10. Contour plots of $f(x, y) = (x-a)^2 + (y-b)^2 + 0.1 \sin(2\pi x) \cos(2\pi y)$ for different values of a . When $a = 0$, f has a single global minimum (circular contours); for $a = 0.25$ and $a = 0.5$, the perturbation introduces multiple local minima.

is no longer globally contractive. Near the global minimum (a, b) , $\nabla^2 f \approx 2I$ remains positive definite and smooth, so there exists a neighborhood $\mathcal{N}(a, b)$ where

$$\|T(z_1) - T(z_2)\| \leq c\|z_1 - z_2\|, \quad c < 1,$$

which means that the map is locally a contraction. However, away from this neighborhood, the Hessian may lose positive definiteness and the mapping can expand rather than contract.

Thus, for $\varepsilon > 0$, Newton's method is locally a contraction near the global minimum but not globally contractive under some certain values of a , because the perturbation $\sin(2\pi x) \cos(2\pi y)$ creates multiple local minima and saddles that break global convexity.

(g) Contraction condition for gradient descent under perturbation. In one dimension, a mapping $T : \mathbb{R} \rightarrow \mathbb{R}$ is a contraction if there exists $0 < c < 1$ such that $|T(x_1) - T(x_2)| \leq c|x_1 - x_2|$. When T is differentiable, this is equivalent to $|T'(x)| < 1$ for all x , since the derivative measures the local stretching factor.

In multiple dimensions, $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is differentiable and the role of the derivative is replaced by the *Jacobian matrix* $J_T(x) = \frac{\partial T(x)}{\partial x^\top}$. Locally, for a small perturbation δ , we have

$$T(x + \delta) - T(x) \approx J_T(x) \delta,$$

so the maximum local stretching factor is given by the operator norm $\|J_T(x)\|_2$. Therefore, the multidimensional analogue of the contraction condition is

$$\|J_T(x)\|_2 < 1 \quad \text{for all } x$$

The gradient-descent iteration is

$$T_\eta(x, y) = (x, y) - \eta \nabla f(x, y),$$

and its Jacobian is

$$J_T(x, y) = I_2 - \eta \nabla^2 f(x, y).$$

A differentiable map T_η is a contraction in the Euclidean norm if

$$\|J_T(x, y)\|_2 < 1 \quad \text{for all } (x, y),$$

which is equivalent to

$$|1 - \eta \lambda_i(\nabla^2 f(x, y))| < 1 \quad \text{for every eigenvalue } \lambda_i(\nabla^2 f(x, y)).$$

The oscillatory term $0.4\pi^2 \sin(2\pi x) \cos(2\pi y)$ varies between ± 3.95 , so the eigenvalues of $\nabla^2 f(x, y)$ roughly lie in

$$\lambda_i(\nabla^2 f) \in [2 - 3.95, 2 + 3.95] = [-1.95, 5.95].$$

The most restrictive requirement comes from the largest positive eigenvalue $\lambda_{\max} \approx 5.95$, which gives the bound

$$0 < \eta < \frac{2}{\lambda_{\max}} \approx \frac{2}{5.95} \approx 0.34.$$

Thus, for all step sizes satisfying $0 < \eta < 0.34$, $|1 - \eta\lambda_i| < 1$ holds for all possible eigenvalues of $\nabla^2 f(x, y)$, and hence T_η is a contraction mapping. If η exceeds this range, the largest-eigenvalue direction will be amplified, and the iteration ceases to be contractive. And in our case, $\epsilon = 0.1$, so gradient descent is a contraction mapping.